



Medical Image Segmentation: A Review of Modern Architectures

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Abstract. Medical image segmentation involves identifying regions of interest in medical images. In modern times, there is a great need to develop robust computer vision algorithms to perform this task in order to reduce the time and cost of diagnosis and thus to aid quicker prevention and treatment of a variety of diseases. The approaches presented so far, mainly follow the U-type architecture proposed along with the UNet model, they implement encoder-decoder type architectures with fully convolutional networks, and also transformer architectures, exploiting both attention mechanisms and residual learning, and emphasizing information gathering at different resolution scales. Many of these architectural variants achieve significant improvements in quantitative and qualitative results in comparison to the pioneer UNet, while some fail to outperform it. In this work, 11 models designed for medical image segmentation, as well as other types of segmentation, are trained, tested and evaluated on specific evaluation metrics, on four publicly available datasets related to gastric polyps and cell nuclei, which are first augmented to increase their size in an attempt to address the problem of the lack of a large amount of medical data. In addition, their generalizability and the effect of data augmentation on the scores of the experiments are also examined. Finally, conclusions on the performance of the models are provided and future extensions that can improve their performance in the task of medical image segmentation are discussed.

Keywords: Medical image segmentation · Polyps · Cell nuclei · UNet · Vnet · Attention UNet · TransUNet · SwinUNet · ResUNet · ResUNet++ · R2U-Net · MSRF-Net · DeepLabv3+ · ResUNet-a · CVC-ClinicDB · Kvasir-SEG · 2018 Data Science Bowl · SegPC

1 Introduction

Medical image segmentation is an essential process for faster and better diagnosis of diseases and for ensuring faster and more effective treatment of a large number of patients simultaneously, reducing the need for human intervention as well as limiting human errors, time and cost. It belongs to the field of semantic segmentation and involves the identification of regions of interest in medical data

such as 2D images, 3D brain MRIs [26, 42, 46], CT scans [27, 28], ultrasounds and X-rays [31], among others. Segmentation results can help identify areas associated with diseases, such as gastrointestinal polyps, in order to analyze them and decide if they are cancerous and need to be removed. This diagnosis can be life-critical for patients as it can help prevent serious forms of cancer. Due to the slow and tedious process of manual segmentation [43], there is a great need for computer vision algorithms that can perform segmentation quickly and accurately without the requirement for human involvement. Other challenges in the field of medical imaging are the small amount of available data, and the need for professionals specialized in this field when generating the data. When annotating polyps in images for example, protocols and guidelines are defined by the expert performing the annotation at that time. However, there are discrepancies between experts, for example in deciding whether a region is cancerous or not. Furthermore, due to the quite significant variation between data due to different imaging quality, the software used to perform the annotation, the expert performing the annotation, the brightness of the images and general biological variations, segmentation algorithms need to be robust and have high generalizability.

In this paper we perform a thorough study of the most important architectures presented in the field of medical image segmentation, we test and evaluate them on four publicly available and widely used datasets and we compare the results to draw conclusions on which architectures are most favorable to the task at hand and which are most promising for future developments. The process followed is illustrated in Fig. 1. Specifically, after the data were collected, split into training, test and validation sets, they were preprocessed and augmented to increase their size and introduce an additional diversity factor. The 11 models, discussed in a following section, are then trained on the augmented data, on the non augmented data and on supersets resulting from combining sets, with appropriately selected hyperparameters and loss functions. The models are then tested over the test subsets, the segmentation masks are generated and qualitatively compared with the actual masks, and the selected evaluation metrics are calculated. Once all the data are collected, a qualitative and quantitative comparison of the architectures follows. In addition, experiments and comparisons are performed to test the effect of data augmentation on the performance of the models, as well as testing the generalization ability of the models through training and evaluation on different datasets.

2 Related Work

In recent years, Convolutional Neural Networks (CNN) have dominated the field of segmentation in various types of medical data, but also in general [6, 29, 30]. The most modern architectures for semantic and instance segmentation are usually encoder-decoder networks, whose success is due to skip-connections, which allow semantically important and dense feature maps to be forwarded from the encoder to the decoder and reduce the semantic gap between the two subnetworks. The implementation parameters of this type of models differ depending

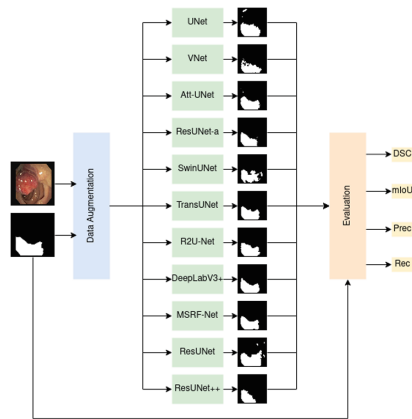


Fig. 1. Entire work pipeline of the essay

on the biomedical application. Moreover, the small amount of publicly available data makes the models' task more difficult and requires specialized design in order to be addressed as a problem. One of the first and most popular approaches to the task of semantic medical image segmentation that addresses the above problem is UNet [39], which converted the Fully Convolutional Network (FCN) of [32], which has only convolutional layers, into a U-type encoder-decoder architecture in which low- and high-level features are combined via skip-connections. However, a semantic gap between these features may occur. In order to bridge the gap, it is proposed in [22] to add convolutional units to the skip-connections. In [35], a version of UNet with an attention mechanism is proposed which helps to implement a pruning mechanism to skip unnecessary spatial features passing through the skip-connections. UNet was not only limited to the medical image domain but versions of UNet for different types of segmentation appeared, such as [13]. The 3D version of UNet, VNet [34] performs segmentation on sparsely annotated volumetric images, and consists of an FCN with skip-connections. Residual links improve information flow while reducing the convergence time, which is why they were preferred in several approaches, such as in [49]. In [25], Squeeze and Excitation blocks, attention mechanisms as well as Atrous Spatial Pyramid Pooling (ASPP) are added to the ResUNet model. In [2], residual connections are combined into a recurrent network which leads to better feature representations. The ASPP mechanism appeared in [9] to pool global features, and evolved in [10] into an architecture that uses skip-connections between the encoder and decoder. Squeeze and Excitation blocks were proposed in [20]. These blocks model interdependencies between channels and extract global maps that emphasize important features and ignore less relevant features. The concept of gated shape stream appeared in [44] and served to produce more detailed segmentation maps by taking into account the shape of the region of interest. To maintain high resolution representations, it was proposed to exchange low and

high level features between different resolution scales, which is shown in [45] to lead to more spatially accurate final segmentation maps. Finally, in [41] many of the above techniques such as the Squeeze and Excitation blocks, the gated shape stream, the attention mechanism and multi-scale fusion are combined into a robust model that consistently performs on the types of medical data contained in the present work.

3 Experimental Setup

3.1 Datasets

In this section we present the datasets used for the training, the validation and the testing process of the models. Unfortunately, there are not enough public datasets of sufficient size and quality for the task of medical image segmentation. To address the first problem of insufficient sample number, data augmentation works as a catalyst. However, the second problem, that of quality, is not considered an easy challenge to address because the quality of the images can only be improved when there is a significant breakthrough in the sample acquisition technology. The sets we used belong to two broad categories involving polyp segmentation and cell nuclei segmentation respectively. Different types of sets were chosen in order to test the robustness and the generalizability of the models. The numbers of samples of each dataset is shown in Table 1(a), and examples from each dataset are shown in Fig. 2.

One of the datasets used in this paper is CVC-ClinicDB [4], which is an open access database of colonoscopy video frames that contain various examples of polyps. The dataset consists of images with a resolution of 384×288 pixels and their binary segmentation masks corresponding to the areas covered by polyps in the images. The images are property of Hospital Clinic, Barcelona, Spain and the masks are property of the Computer Vision Center, Barcelona, Spain.

Another dataset used in this work is Kvasir-SEG [24] which is an open access dataset with images of gastrointestinal polyps and corresponding segmentation masks, with manual annotation and verification by an experienced gastroenterologist. The resolution of the images varies from 332×487 to 1920×1072 pixels. The image and the mask files are encoded with JPEG compression. Data were collected using endoscopic equipment at Vestre Viken Health Trust (VV) in Norway.

The 2018 Data Science Bowl dataset [5] contains a large number of segmented images of cell nuclei. The images were acquired under a variety of conditions and vary in terms of cell type, magnification and imaging mode. Each segmentation mask contains one nucleus, so each image can have more than one mask assigned to it. This dataset contains images from more than 30 biological experiments on different samples, cell strains, microscopy instruments, imaging conditions, operators, research facilities and staining protocols. The annotations were manually created by a team of expert biologists at the Broad Institute.

For the SegPC dataset [14, 17–19], microscopic images were recorded from bone marrow aspirate slides of patients diagnosed with Multiple Myeloma, a type

of white blood cancer. The slides were stained and plasma cells, which are cells of interest, were segmented. Images were recorded in raw BMP format. This dataset aims at a robust cell segmentation which is the first step in creating a computer-assisted diagnostic tool for plasma cell cancer, namely Multiple Myeloma (MM).

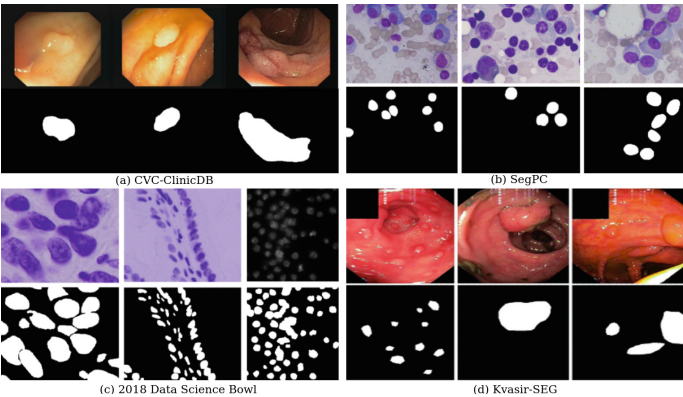


Fig. 2. Examples of image-mask pairs from all datasets [4,5,14,17–19,24]

Some of the datasets were originally not in the format dictated by the experimental procedure, so a preprocessing procedure was needed. In particular, all the datasets were divided into training, validation and test folders in the ratio of 80%, 10% and 10% respectively. For the 2018 Data Science Bowl and SegPC datasets there was additional preprocessing. In the 2018 Data Science Bowl a separate mask is given for each nucleus identified in the image. These masks were merged so that for each image there is a unique binary mask that includes all nuclei. Similarly for the SegPC all the masks corresponding to one image are merged into a final binary mask containing all the nuclei (ignoring the cytoplasm to remain consistent with the 2018 Data Science Bowl dataset). Finally, joining the Kvasir-SEG and CVC-CLinicDB datasets led to the creation of a new larger dataset that was named “Polyps dataset”, and similarly joining the 2018 Data Science Bowl and the SegPC datasets led to the “Cells Dataset”. The creation of these sets was deemed necessary for some experiments to be reported in later sections concerning the generalizability of the models.

Due to the relatively small number of samples that are publicly available for the task of medical image segmentation in the areas considered in this paper, it is necessary to augment the data [37]. Techniques such as random rotation, shear, zoom, horizontal and vertical flip, were applied to the training sets resulting in a large increase in the number of samples, not exceeding the memory and runtime limits of the available equipment used in this work. The updated numbers of samples are shown on Table 1(b).

Table 1. (a) Number of samples in all datasets before augmentation (b) Number of samples in all datasets after augmentation

Dataset	Images Train Valid Test				Dataset	Images Train Valid Test			
CVC-ClinicDB	612	489	61	62	CVC-ClinicDB	860	737(+50%)	61	62
Kvasir-SEG	1000	800	100	100	Kvasir-SEG	1496	1296(+62%)	100	100
2018 Data Science Bowl	670	536	67	67	2018 Data Science Bowl	918	784(+46%)	67	67
SegPC	497	397	50	50	SegPC	745	645(+62%)	50	50
Polyps Dataset	1612	1289	161	162	Polyps Dataset	2356	2033(+58%)	161	162
Cells Dataset	1167	933	117	117	Cells Dataset	1664	1429(+60%)	117	117
(a)					(b)				

3.2 Model Architectures

In this section we present the models and their architectures as used in the experiments of this paper. Several of the most widely used models in the field of medical image segmentation as well as some State-of-the-Art models were chosen, but not only limited to that. We have used open-source code provided by the respective authors for all the models.

UNet’s [39] architecture consists of a contracting path to capture context and a symmetric expanding path that enables precise localization. Such a network can be trained end-to-end from very few images. The contracting path follows the typical architecture of a convolutional network whereas the expansive path consists of upsampling of the feature map, convolutions and a concatenation with the corresponding cropped feature map from the contracting path.

VNet [34] utilises the power of fully convolutional neural networks to process MRI volumes. It also introduces residual connections along with a novel objective function. In this work, a 2D version of VNet is used.

Attention UNet [35] is an attention gate model for medical imaging that can focus on target objects of variable shape and size. It learns to ignore non relevant features and to highlight the ones that are important to a specific task, while also eliminating the need for external localization modules.

ResUNet [49] is a network which combines residual learning with U-Net, and is designed for road area extraction. The network consists of residual units and has the extra advantages of easy training and need for fewer parameters.

ResUNet++ [25] is an improved ResUNet architecture designed for segmentation of colonoscopic images, it takes advantage of residual blocks, squeeze and excitation blocks, Atrous Spatial Pyramidal Pooling (ASPP) and attention blocks.

ResUNet-a [13] is a deep learning architecture designed for the semantic segmentation of high resolution aerial images, that uses a UNet backbone along with residual connections, atrous convolutions, pyramid scene parsing pooling and multi-tasking inference.

R2U-Net [2] is a Recurrent Residual Convolutional Neural Network inspired by UNet, which leverages the power of UNet, residual networks, as well as Recurrent CNNs. It succeeds in easy training and more effective feature representation for segmentation tasks.

SwinUNet [7] is a pure Transformer UNet inspired architecture for medical image segmentation. The tokenized image patches are fed into the Transformer-based U-shaped Encoder-Decoder architecture with skip-connections for semantic feature learning.

TransUNet [8] combines the strengths of both Transformers and U-Net and is presented as a strong alternative for medical image segmentation. The Transformer encodes tokenized image patches from a convolutional neural network (CNN) feature map as the input sequence and the decoder upsamples the encoded features which are then combined with the high-resolution CNN feature maps to enable precise localization.

DeepLabv3+ [10] is a novel architecture which combines the multi-scale fusion advantages of spatial pyramid pooling modules and the sharp object boundaries of encoder-decoder networks. It extends DeepLabv3 [9] by adding a decoder module to refine the segmentation results especially along object boundaries.

MSRF-Net [41] is a Multi-Scale Residual Fusion Network specially designed for medical image segmentation. It is able to exchange multi-scale features of varying receptive fields using Dual-Scale Dense Fusion (DSDF) blocks which can exchange information rigorously across two different resolution scales, allowing the preservation of resolution.

3.3 Metrics

Loss Functions. One of the loss functions used in this work is Binary Cross Entropy which calculates the performance of a binary classification model (the type of semantic segmentation in this task is equivalent to classification with two classes, 0 and 1) that produces as output values of probability 0 to 1. It is described by Formula 1.

$$\mathcal{L}_{BCE} = (1 - y)\log(1 - \hat{y}) + y\log\hat{y} \quad (1)$$

where y is the real value and $\hat{y}$ is the predicted value. In addition to Binary Cross Entropy, there is also Dice Loss which is a measure of the overlap between the real image and the model-computed image and is described by Formula 2.

$$\mathcal{L}_{DSC} = 1 - \frac{2y\hat{y} + 1}{y + \hat{y} + 1} \quad (2)$$

where y is the real value and $\hat{y}$ is the predicted value. In all of the models presented above, both loss functions were used in example experiments and for each one of them, the loss function that led to the best results was eventually selected. In the MSRF-Net model a combination of loss functions defined in Formula 3 was used:

$$\mathcal{L}_{MSRF} = \mathcal{L}_{comb} + \mathcal{L}_{comb}^{DS^0} + \mathcal{L}_{comb}^{DS^1} + \mathcal{L}_{BCE}^{SS} \quad (3)$$

where $\mathcal{L}_{comb} = \mathcal{L}_{DSC} + \mathcal{L}_{BCE}$ is the combination of Dice and Binary Cross Entropy losses, $\mathcal{L}_{comb}^{DS^0}$ and $\mathcal{L}_{comb}^{DS^1}$ are the two deep supervision output losses of the model, and $\mathcal{L}_{BCE}^{SS}$ is the Binary Cross Entropy Loss computed for the shape stream of the MSRF-Net model [41].

Evaluation Metrics. For the evaluation of the models’ performance we used standard medical image segmentation evaluation metrics, such as the (mean) Dice Coefficient described in Formula 4 and (mean) Intersection Over Union described in Formula 5, which measure the overlap between the predicted and the true images, as well as Precision described in Formula 6 and Recall described in Formula 7, which measure the ratio of correct predictions in similar ways.

$$DSC = \frac{2|A \cap B|}{|A| + |B|} \quad (4)$$

$$IoU = \frac{|A \cap B|}{|A \cup B|} \quad (5)$$

$$Precision = \frac{TruePositives}{TruePositives + FalsePositives} \quad (6)$$

$$Recall = \frac{TruePositives}{TruePositives + FalseNegatives} \quad (7)$$

where $|A|$ and $|B|$ are the sets of the activations in the predicted mask and the ground truth mask respectively.

3.4 Implementation Details

All the augmentation, training and evaluation code has been implemented using the Keras [11] framework with TensorFlow [1] as the backend. All experiments were conducted on ARIS [16], the Greek supercomputer developed and maintained by GRNET in Athens, and more specifically on 2 x NVIDIA Tesla k40m accelerated GPU nodes. All the images and masks from the four datasets were resized to 256×256 , the number of epochs is set to 200. The optimizer for all models was chosen to be Adam, with a learning rate of 0.0001. We used a batch size of 8 for CVC-ClinicDB and SegPC and a batch size of 16 for Kvasir-SEG and 2018 Data Science Bowl.

4 Results

Table 2(a) shows the results of the experiments on the CVC-ClinicDB dataset which, as previously analysed, includes images of polyps derived from colonoscopy videos. The best performing model in terms of Dice Coefficient and mean Intersection over Union metrics is DeepLabv3+ followed by MSRF-Net, R2U-net, TransUnet and UNet. The MSRF-Net and R2U-Net achieve the highest Precision, while the highest Recall is achieved by DeepLabv3+. The good performance of these models is visually confirmed by Fig. 3 and the masks produced by them resemble the real masks to a good extent. Models such as SwinUNet, VNet and ResUNet appear to fail on this particular dataset. The masks generated by them show significant imperfections compared to the real masks.

Table 2(b) shows the results of the experiments on the Kvasir-SEG dataset which also includes images of gastrointestinal polyps. The most robust models

Table 2. Quantitative results of our experiments on the four original augmented datasets

Method	DSC	mIoU	Precision	Recall	Method	DSC	mIoU	Precision	Recall
MSRF-Net	0.87	0.84	0.95	0.89	MSRF-Net	0.86	0.73	0.89	0.80
UNet	0.92	0.80	0.91	0.87	UNet	0.90	0.71	0.89	0.78
VNet	0.81	0.64	0.79	0.76	VNet	0.76	0.49	0.73	0.60
Att-UNet	0.87	0.69	0.92	0.73	Att-UNet	0.86	0.62	0.82	0.72
ResUNet-a	0.86	0.71	0.85	0.81	ResUNet-a	0.78	0.49	0.73	0.60
SwinUNet	0.79	0.56	0.70	0.71	SwinUNet	0.77	0.50	0.79	0.66
TransUNet	0.94	0.80	0.92	0.86	TransUNet	0.90	0.70	0.87	0.79
R2U-Net	0.94	0.83	0.95	0.88	R2U-Net	0.88	0.69	0.85	0.79
DeepLabv3+	0.99	0.89	0.94	0.94	DeepLabv3+	0.96	0.73	0.87	0.82
ResUNet	0.70	0.64	0.80	0.77	ResUNet	0.60	0.49	0.66	0.66
ResUNet++	0.78	0.72	0.90	0.78	ResUNet++	0.65	0.55	0.74	0.69

(a) CVC-ClinicDB					(b) Kvast-SEG				
Method	DSC	mIoU	Precision	Recall	Method	DSC	mIoU	Precision	Recall
MSRF-Net	0.91	0.86	0.91	0.95	MSRF-Net	0.80	0.68	0.82	0.79
UNet	0.95	0.87	0.92	0.94	UNet	0.88	0.66	0.81	0.78
VNet	0.94	0.85	0.91	0.93	VNet	0.85	0.57	0.75	0.71
Att-UNet	0.95	0.86	0.92	0.93	Att-UNet	0.89	0.66	0.80	0.79
ResUNet-a	0.91	0.76	0.90	0.83	ResUNet-a	0.77	0.41	0.60	0.57
SwinUNet	0.94	0.85	0.92	0.92	SwinUNet	0.80	0.45	0.63	0.62
TransUNet	0.95	0.87	0.93	0.93	TransUNet	0.90	0.68	0.81	0.80
R2U-Net	0.95	0.86	0.90	0.95	R2U-Net	0.89	0.69	0.85	0.78
DeepLabv3+	0.98	0.85	0.91	0.93	DeepLabv3+	0.98	0.64	0.78	0.78
ResUNet	0.88	0.84	0.89	0.94	ResUNet	0.77	0.64	0.74	0.82
ResUNet++	0.89	0.83	0.89	0.93	ResUNet++	0.76	0.62	0.76	0.77

(c) 2018 Data Science Bowl					(d) SegPC				
Method	DSC	mIoU	Precision	Recall	Method	DSC	mIoU	Precision	Recall
MSRF-Net	0.91	0.86	0.91	0.95	MSRF-Net	0.80	0.68	0.82	0.79
UNet	0.95	0.87	0.92	0.94	UNet	0.88	0.66	0.81	0.78
VNet	0.94	0.85	0.91	0.93	VNet	0.85	0.57	0.75	0.71
Att-UNet	0.95	0.86	0.92	0.93	Att-UNet	0.89	0.66	0.80	0.79
ResUNet-a	0.91	0.76	0.90	0.83	ResUNet-a	0.77	0.41	0.60	0.57
SwinUNet	0.94	0.85	0.92	0.92	SwinUNet	0.80	0.45	0.63	0.62
TransUNet	0.95	0.87	0.93	0.93	TransUNet	0.90	0.68	0.81	0.80
R2U-Net	0.95	0.86	0.90	0.95	R2U-Net	0.89	0.69	0.85	0.78
DeepLabv3+	0.98	0.85	0.91	0.93	DeepLabv3+	0.98	0.64	0.78	0.78
ResUNet	0.88	0.84	0.89	0.94	ResUNet	0.77	0.64	0.74	0.82
ResUNet++	0.89	0.83	0.89	0.93	ResUNet++	0.76	0.62	0.76	0.77

with respect to the mIoU metric are MSRF-Net and DeepLabv3+ followed by UNet and TransUNet, while with respect to the DSC metric the most robust model is DeepLabv3+ followed by R2U-Net, TransUNet and UNet. The highest Precision is achieved by MSRF-Net and UNet, and the highest Recall is achieved by DeepLabv3+. Observing the visual results in Fig. 3, the strong models produce fairly qualitative masks, in which the target polyp is detected, however, additional small regions may be misidentified as polyps. The models that seem to fail are VNet, ResUNet-a and SwinUNet, which is visually confirmed by the inaccurate generated masks.

Table 2(c) lists the results on the 2018 Data Science Bowl dataset which includes images of nuclei of different cell types. An initial observation is that all models achieve significantly higher performance compared to the other datasets. According to the DSC metric the strongest models are DeepLabv3+, R2U-Net, TransUNet, Attention UNet, SwinUNet, UNet and VNet while according to mIoU the strongest are TransUNet, UNet followed by MSRF-Net, Attention UNet, R2U-Net, SwinUNet, VNet and DeepLabv3+. The highest Precision is achieved by TransUNet and the highest Recall is achieved by MSRF-Net and R2U-Net. The only model that shows significantly lower performance than the others is ResUNet-a. Figure 3 shows the very high similarity between the masks of the models and the actual mask in terms of the grayscale image, while in terms of the color image the masks of the models that achieve the highest scores in the mIoU metric are clearly of higher quality.

Finally, Table 2(d) lists the results on the SegPC dataset which consists of plasma cell images from Multiple Myeloma patients. The model performances for this dataset are lower than the above datasets, which makes SegPC the more challenging dataset. The strongest model based on the DSC metric is DeepLabv3+ followed by TransUNet, R2U-Net, Attention UNet and UNet, while

based on the mIoU metric the strongest model is R2U-Net followed by MSRF-Net and TransUNet. The highest Precision is achieved by the R2U-Net model and the highest Recall is achieved by the ResUNet model. These results are confirmed visually in Fig. 3. The models that appear to fail based on both the quantitative results in Table 2(d) and the visual results in Fig. 3 are SwinUNet and ResUNet-a.

We also examine the effect of the data augmentation on the quantitative results of the experiments. The greater the increase in scores observed with the increasing size of the datasets, the more the power of the models is highlighted, as it is shown that their performance will improve in the event that more and larger medical image datasets are produced in the future.

Table 3 presents the results without data augmentation. It is observed that data augmentation causes an approximate 4–5% increase in the mIoU metric, a 2–3% increase in the DSC metric, a 1–2% increase in the Precision metric and a 3–4% increase in the Recall metric. Some models benefit more from the larger data volume than others, for example DeepLabv3+ achieves a 2% increase in the mIoU metric while MSRF-Net achieves a 4% increase in the Kvasir-SEG dataset. Other models seem to not achieve an increase through data augmentation, for example ResUNet, ResUNet++ in the CVC-ClinicDB dataset. In general, the increasing trend in model performance as a function of increasing data volume is confirmed. Models such as MSRF-Net, R2U-Net, TransUNet seem to scale faster and to a greater extent with increasing data size.

Table 3. Quantitative results of our experiments on the four original non-augmented datasets

Method	DSC	mIoU	Precision	Recall	Method	DSC	mIoU	Precision	Recall
MSRF-Net	0.89	0.83	0.94	0.80	MSRF-Net	0.81	0.69	0.84	0.79
UNet	0.90	0.78	0.91	0.85	UNet	0.87	0.64	0.83	0.74
VNet	0.79	0.59	0.78	0.71	VNet	0.75	0.44	0.69	0.55
Att-UNet	0.83	0.67	0.86	0.75	Att-UNet	0.84	0.58	0.84	0.66
ResUNet-a	0.84	0.70	0.89	0.76	ResUNet-a	0.75	0.46	0.64	0.62
SwinUNet	0.78	0.54	0.70	0.71	SwinUNet	0.75	0.45	0.65	0.60
TransUNet	0.91	0.76	0.90	0.83	TransUNet	0.87	0.66	0.80	0.80
R2U-Net	0.90	0.76	0.92	0.81	R2U-Net	0.87	0.66	0.85	0.75
DeepLabv3+	0.99	0.87	0.96	0.91	DeepLabv3+	0.96	0.75	0.99	0.83
ResUNet	0.66	0.64	0.81	0.76	ResUNet	0.54	0.45	0.58	0.67
ResUNet++	0.77	0.72	0.94	0.76	ResUNet++	0.60	0.50	0.70	0.63
(a) CVC-ClinicDB					(b) Kvasir-SEG				
Method	DSC	mIoU	Precision	Recall	Method	DSC	mIoU	Precision	Recall
MSRF-Net	0.91	0.86	0.91	0.94	MSRF-Net	0.79	0.67	0.81	0.78
UNet	0.95	0.86	0.91	0.95	UNet	0.86	0.63	0.78	0.77
VNet	0.94	0.85	0.91	0.93	VNet	0.84	0.57	0.75	0.71
Att-UNet	0.95	0.86	0.91	0.94	Att-UNet	0.88	0.65	0.78	0.80
ResUNet-a	0.93	0.83	0.88	0.94	ResUNet-a	0.74	0.39	0.52	0.39
SwinUNet	0.94	0.85	0.92	0.92	SwinUNet	0.77	0.43	0.56	0.65
TransUNet	0.95	0.86	0.95	0.90	TransUNet	0.88	0.63	0.75	0.80
R2U-Net	0.95	0.86	0.92	0.93	R2U-Net	0.85	0.63	0.77	0.78
DeepLabv3+	0.98	0.84	0.92	0.91	DeepLabv3+	0.97	0.61	0.76	0.76
ResUNet	0.89	0.85	0.90	0.93	ResUNet	0.70	0.56	0.69	0.76
ResUNet++	0.89	0.84	0.89	0.94	ResUNet++	0.72	0.60	0.79	0.72
(c) 2018 Data Science Bowl					(d) SegPC				

The generalizability is an important characteristic of a model and thus considered necessary to evaluate. In order to evaluate the generalization ability [23], models are trained on one of the two datasets including gastrointestinal polyps (CVC-ClinicDB, Kvasir-SEG) and evaluated on the other dataset. Similarly for

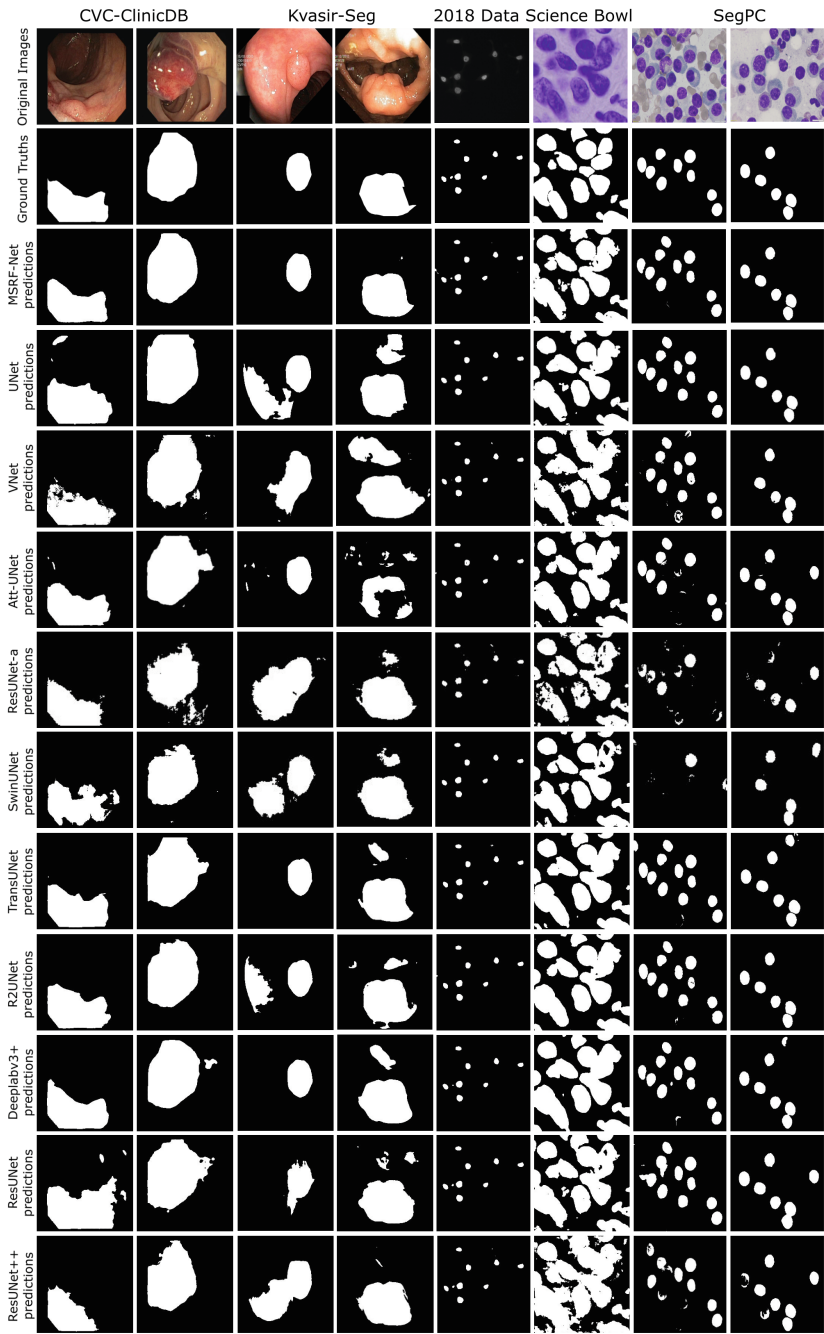


Fig. 3. Qualitative results of our experiments on the four original augmented datasets

the two datasets containing cell nuclei (2018 Data Science Bowl, SegPC). The results of the new experiments are shown in Tables 4(a)–(c). The results of the training experiment on SegPC and evaluation on the 2018 Data Science Bowl are omitted because they are of extremely low quality, making SegPC an inadequate dataset for training, in terms of generalizability.

While the results are generally unsatisfactory, with scores on the mIoU metric not exceeding 53% on the CVC-ClinicDB set, 66% on the Kvasir-SEG set and 30% on the 2018 Data Science Bowl set, the results can be used as a benchmark for the models’ generalizability. The DeepLabv3+ model shows the highest generalization ability by achieving the highest scores in DCS, mIoU, while in contrast the ResUNet-a model fails in this area. Models such as MSRF-Net, VNet, TransUNet, R2U-Net, Attention UNet, UNet and ResUNet++ show quite good results in these experiments, therefore it is judged that they can generalize to a sufficient degree. An additional observation is that models trained on Kvasir-SEG perform better on CVC-ClinicDB while the opposite is not true.

Table 4. Quantitative generalizability results of our experiments on all six datasets

Method	DSC	mIoU	Precision	Recall	Method	DSC	mIoU	Precision	Recall	Method	DSC	mIoU	Precision	Recall
MSRF-Net	0.41	0.30	0.42	0.52	MSRF-Net	0.52	0.42	0.53	0.67	MSRF-Net	0.42	0.27	0.27	0.96
UNet	0.53	0.23	0.24	0.84	UNet	0.75	0.45	0.69	0.56	UNet	0.69	0.29	0.29	0.94
VNet	0.59	0.26	0.28	0.80	VNet	0.66	0.30	0.44	0.49	VNet	0.69	0.29	0.30	0.93
Att-UNet	0.56	0.24	0.26	0.80	Att-UNet	0.72	0.42	0.78	0.47	Att-UNet	0.68	0.28	0.29	0.95
ResUNet-a	0.46	0.10	0.25	0.10	ResUNet-a	0.55	0.14	0.19	0.38	ResUNet-a	0.54	0.11	0.23	0.14
SwinUNet	0.55	0.24	0.25	0.76	SwinUNet	0.63	0.26	0.30	0.63	SwinUNet	0.66	0.26	0.27	0.95
TransUNet	0.57	0.25	0.27	0.82	TransUNet	0.76	0.47	0.70	0.59	TransUNet	0.68	0.28	0.28	0.96
R2U-Net	0.55	0.24	0.25	0.78	R2U-Net	0.74	0.46	0.77	0.53	R2U-Net	0.66	0.26	0.27	0.97
DeepLabv3+	0.91	0.53	0.66	0.72	DeepLabv3+	0.97	0.66	0.93	0.70	DeepLabv3+	0.87	0.28	0.29	0.94
ResUNet	0.35	0.22	0.23	0.85	ResUNet	0.38	0.30	0.50	0.43	ResUNet	0.39	0.25	0.25	0.97
ResUNet++	0.30	0.23	0.34	0.40	ResUNet++	0.51	0.37	0.52	0.57	ResUNet++	0.45	0.30	0.30	0.96

(a) Trained on CVC-ClinicDB Tested on Kvasir-SEG					(b) Trained on Kvasir-SEG Tested on CVC-ClinicDB					(c) Trained on 2018 Data Science Bowl Tested on SegPC				
Tested on	CVC-ClinicDB				Tested on	Kvasir-SEG				Tested on	2018 Data Science Bowl			
Method	DSC	mIoU	Precision	Recall	Method	DSC	mIoU	Precision	Recall	Method	DSC	mIoU	Precision	Recall
MSRF-Net	0.90	0.86	0.94	0.91	0.85	0.75	0.88	0.84	0.84	MSRF-Net	0.91	0.87	0.93	0.93
UNet	0.90	0.77	0.94	0.80	0.89	0.69	0.88	0.76	0.76	UNet	0.95	0.87	0.93	0.93
VNet	0.78	0.54	0.77	0.64	0.83	0.58	0.82	0.66	0.66	VNet	0.92	0.80	0.94	0.84
Att-UNet	0.91	0.74	0.94	0.78	0.87	0.64	0.82	0.75	0.75	Att-UNet	0.94	0.86	0.94	0.91
ResUNet-a	0.81	0.61	0.92	0.65	0.79	0.53	0.77	0.63	0.63	ResUNet-a	0.94	0.84	0.91	0.92
SwinUNet	0.78	0.48	0.66	0.64	0.75	0.44	0.56	0.68	0.68	SwinUNet	0.93	0.82	0.90	0.91
TransUNet	0.92	0.74	0.88	0.82	0.89	0.69	0.83	0.81	0.81	TransUNet	0.95	0.87	0.94	0.93
R2U-Net	0.89	0.75	0.94	0.79	0.90	0.73	0.92	0.78	0.78	R2U-Net	0.94	0.87	0.92	0.93
DeepLabv3+	0.99	0.86	0.93	0.92	0.96	0.76	0.86	0.87	0.87	DeepLabv3+	0.98	0.85	0.92	0.91
ResUNet	0.55	0.50	0.38	0.77	0.58	0.46	0.54	0.75	0.75	ResUNet	0.85	0.72	0.90	0.78
ResUNet++	0.76	0.68	0.93	0.71	0.68	0.59	0.83	0.68	0.68	ResUNet++	0.86	0.74	0.82	0.80

(d) Trained on Polyps Dataset Tested on CVC-ClinicDB and Kvasir-SEG					(e) Trained on Cells Dataset Tested on 2018 Data Science Bowl and SegPC				
Tested on	CVC-ClinicDB				Tested on	2018 Data Science Bowl			
Method	DSC	mIoU	Precision	Recall	Method	DSC	mIoU	Precision	Recall
MSRF-Net	0.90	0.86	0.94	0.91	0.85	0.75	0.88	0.84	0.84
UNet	0.90	0.77	0.94	0.80	0.89	0.69	0.88	0.76	0.76
VNet	0.78	0.54	0.77	0.64	0.83	0.58	0.82	0.66	0.66
Att-UNet	0.91	0.74	0.94	0.78	0.87	0.64	0.82	0.75	0.75
ResUNet-a	0.81	0.61	0.92	0.65	0.79	0.53	0.77	0.63	0.63
SwinUNet	0.78	0.48	0.66	0.64	0.75	0.44	0.56	0.68	0.68
TransUNet	0.92	0.74	0.88	0.82	0.89	0.69	0.83	0.81	0.81
R2U-Net	0.89	0.75	0.94	0.79	0.90	0.73	0.92	0.78	0.78
DeepLabv3+	0.99	0.86	0.93	0.92	0.96	0.76	0.86	0.87	0.87
ResUNet	0.55	0.50	0.38	0.77	0.58	0.46	0.54	0.75	0.75
ResUNet++	0.76	0.68	0.93	0.71	0.68	0.59	0.83	0.68	0.68

In addition, in the experiments concerning the larger datasets, the models were trained on the new Polyps, Cells datasets and tested and evaluated on their constituent subsets. The results are shown in Table 4(d)–(e). In these experiments, the suitability of the Polyps and Cells datasets for training and their effect on improving the results is also examined in parallel. According to Table 4(d) and (e), it is shown that training most models on the Polyps set leads to significantly increased scores when evaluating on the CVC-ClinicDB set and less increased scores when evaluating on the Kvasir-SEG set. In contrast, training the models on the Cells set leads to reduced scores when evaluating on the 2018 Data Science Bowl and SegPC sets. Models that respond positively to training on the Polyps and Cells sets compared to training on the individual sets

that comprise them have higher generalizability. As demonstrated in the above experiments, models such as DeepLabv3+, MSRF-Net, VNet, Attention UNet, TransUNet, UNet, and R2U-Net generalize better than the other models.

5 Discussion

Models such as ResUNet and ResUNet-a, designed for domains other than medical image segmentation, do not perform well enough on datasets involving polyps but achieve satisfactory scores on datasets with cell nuclei, indicating that this project is quite specialized and thus requires specialized models to achieve state-of-the-art results. The utilization of residual connections in combination with recurrent networks is found to be very effective in the R2U-Net while their combination with Squeeze and excitation blocks, Atrous Spatial Pyramid Pooling [9, 15, 33, 36, 40] and attention does not lead to any improvement over UNet and at the same time complicates the architecture a lot. The utilization of the attention mechanism does not perform on the polyp datasets however it leads to better identification of cells with disease in the SegPC set and their separation from healthy cells, as is the case with Attention UNet. The appearance of transformers in U-type architectures, while not promising as an idea at first because by their nature transformers lack localization capability, in combination with convolutional networks, as is the case in TransUNet, they are very efficient encoders bringing a small improvement in UNet's scores. In contrast, without using convolutional networks together with transformers, the final results are insufficient, as is the case with SwinUNet. Finally, models that focus more on combining information from different scales and therefore manage to reduce the semantic gap between the low-level encoder features and the high-level decoder features, such as DeepLabv3+ and MSRF-Net, seem to achieve significant improvement in scores on datasets with polyps that are considered more challenging, as well as SegPC. It is important to note that the MSRF-Net model is considerably more demanding to train in terms of computational resources and time than the other models that perform equally well. Moreover, DeepLabv3+'s ResNet50 backbone which is pretrained on the large ImageNet [12] dataset makes it more powerful and contributes to its fast training.

In realistic scenarios where these models would perform polyp segmentation on real images from different laboratories, with different technologies, from different patients, the generalizability is considered an essential feature [41]. The models must be able to perform adequately regardless of parameters such as the equipment with which the images were captured or the patient whom the images were obtained from. It was shown from the study in the previous section that the models with the highest scores also have a high generalization ability.

As for the datasets, those related to polyps are challenging and carry some peculiarities, such as some black boxes that show high contrast in the images and can confuse the models as to the important areas of the image, and little general contrast in the images especially in the polyp locations which makes the task of detecting polyps more difficult. In the datasets associated with cell

nuclei the task of identifying nuclei is considered to be quite easy because there is considerable contrast between them and the plain monochrome background, with difficulties being encountered in identifying specific cells affected by Multiple Myeloma (in SegPC) and in images where nuclei are depicted more faintly or there is significant overlap between them.

6 Conclusions

The models tested in the experiments of this work generally follow the U-architecture with variations such as the use of transformers, the existence of an attention mechanism, the exploitation of residual learning and the exchange of features at different scales. While the original model that inspired the creation of almost all the others, UNet, seems to consistently perform quite well on all datasets, some of the variations presented lead to improved results while others are not as successful. We observe that multi-scale feature fusion via atrous spatial pyramid pooling or information exchange between scales, along with residual connections and attention lead to the most noticeable improvements. The combination of transformers with convolutional networks is also proven to be a promising approach. Data augmentation is beneficial to all the favoured proposed architectures.

While several models with different architectures were considered in this paper and therefore led to a global study of the medical image segmentation task, there are additional models that could be tested, such as DCSAU-Net [48], MCGU-Net [3] but also UNet++ [50], U^2 -Net [38], UNet3+ [21], which add new details and changes that can significantly favor the results of the experiments.

Moreover, in Fig. 3 it is easily observed that in some examples in the CVC-ClinicDB and Kvasir-SEG datasets, the more efficient models produce masks that misidentify as polyps, regions that are not polyps which can reduce their scores. Combining the above models with bounding boxes from a sufficiently powerful Object Detection model, such as Detectron2 [47], either during the evaluation or during training, could lead to the elimination of the incorrectly detected regions.

Finally, a meticulous ablation study could be conducted focused on specific aspects of the mentioned architectures so that more detailed conclusions can be drawn. Also a larger and perhaps more intense data augmentation could certainly bring about an improvement in the results, since an increasing trend in scores as a function of data volume was already observed.

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